

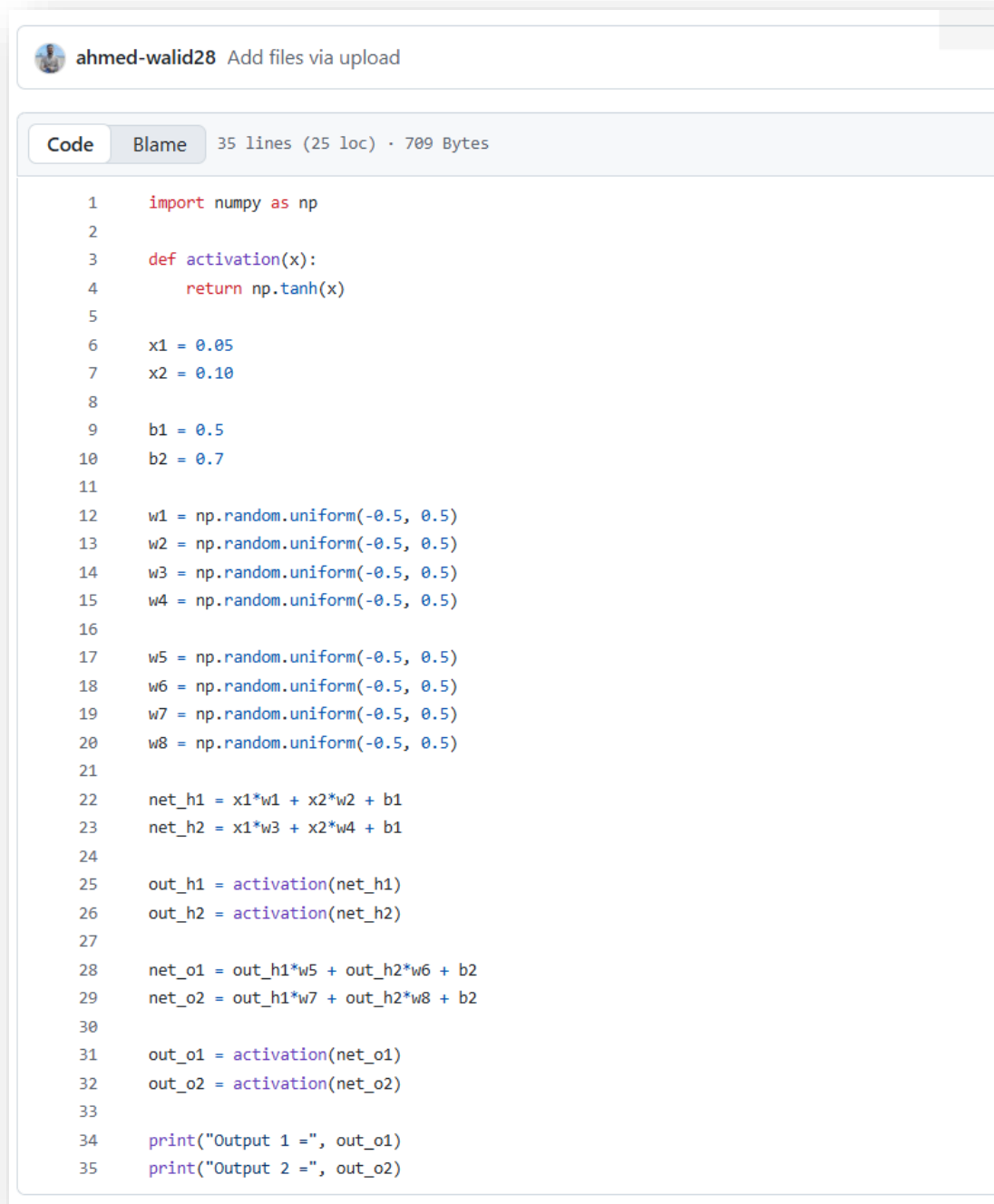
Assignment

1

Student Name
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<https://github.com/ahmed-walid28/ANN-Tanh-Activation-Function/>



The screenshot shows a GitHub repository interface for a user named 'ahmed-walid28'. The repository is titled 'ANN-Tanh-Activation-Function/'. The 'Code' tab is selected, showing a Python script with 35 lines of code. The script defines a 'tanh' activation function and implements a simple neural network with two hidden layers and two output nodes. The network is initialized with random weights and biases. The input values are 0.05 and 0.10. The output values are calculated and printed.

```
1  import numpy as np
2
3  def tanh(x):
4      return np.tanh(x)
5
6  x1 = 0.05
7  x2 = 0.10
8
9  b1 = 0.5
10 b2 = 0.7
11
12 w1 = np.random.uniform(-0.5, 0.5)
13 w2 = np.random.uniform(-0.5, 0.5)
14 w3 = np.random.uniform(-0.5, 0.5)
15 w4 = np.random.uniform(-0.5, 0.5)
16
17 w5 = np.random.uniform(-0.5, 0.5)
18 w6 = np.random.uniform(-0.5, 0.5)
19 w7 = np.random.uniform(-0.5, 0.5)
20 w8 = np.random.uniform(-0.5, 0.5)
21
22 net_h1 = x1*w1 + x2*w2 + b1
23 net_h2 = x1*w3 + x2*w4 + b1
24
25 out_h1 = tanh(net_h1)
26 out_h2 = tanh(net_h2)
27
28 net_o1 = out_h1*w5 + out_h2*w6 + b2
29 net_o2 = out_h1*w7 + out_h2*w8 + b2
30
31 out_o1 = tanh(net_o1)
32 out_o2 = tanh(net_o2)
33
34 print("Output 1 =", out_o1)
35 print("Output 2 =", out_o2)
```